ADVANCED PROBLEMS FOR LECTURE 19

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

(1) These differentials seem like part of the integral notation that we have been working with. Is there a nice interpretation of the integral as a sum of differentials?

(2) Can we make the idea of approximation more strict somehow? E.g. can we bound the error between our function and its tangent approximation in a meaningful way? (Hint: think second derivative)

(3) You will come across this idea in fuller detail in your next course, but instead of approximating with a tangent line, we can consider approximating with a tangent parabola, perhaps. Come up with an appropriate "tangent parabola" equation for a function which is twice-differentiable, and see if you can extend the approximation condition from the previous problem to this case.

(4) Extend your ideas to a degree n polynomial approximation of a function f at a, which is n-times differentiable.

(5) What would happen if the tangent polynomials for f for every degree were 0? Does this imply that f itself is zero?

If you want to learn more about these polynomial approximations, these are named after the mathematician Brook Taylor, and are often called Taylor approximations, Taylor polynomials, or Taylor series (in the infinite series case).